

Pranav Annapareddi

IAM Software Engineer

Experience developing secure backend systems, identity and access management solutions, and contributing to defense-focused projects.

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Active Secret Clearance

TECHNICAL SKILLS:

Languages: Java, Python, C/C++, SQL, HTML, Rust

Skills: Scripts, Unix/Linux, Rest APIs, Computer Networking, AI/ML, Information Security, Network/System Security

Tools: SailPoint, Gitlab, MySQL, Virtual Machines, Docker, CLI, Objdump, Pwntools, GDB, Ropper, radare2, Angr

PROFESSIONAL EXPERIENCE

Deloitte, San Jose, CA - IAM Software Engineer

August 2022 - Current

- Developed secure, scalable RESTful APIs in Java, enforcing IAM policies for over **5 million users**.
- Crafted a security automation testing script simulating client behavior for users by utilizing multi-threading libraries.
- Automated our testing environments by creating Python scripts that validated our API endpoints, significantly increasing coverage and efficiency by identifying edge cases and reducing manual testing efforts by 90%.
- Designed and developed a custom webpage using HTML and CSS, hosting our API information. This centralized page provides a comprehensive API documentation and token management system. Incorporated input validation on executable API calls on the webpage.
- Orchestrated IAM solutions using SailPoint, resulting in a **53% increase in client compliance scores** by elevating security and adherence to regulatory standards such as separation of duties (SOD).

Juniper Networks, San Jose, CA - Software Engineer Intern

May 2020 - August 2020

- Led the artificial intelligence driven solutions team to produce a predictive auto-bandwidth solution for our clients. The predictive model achieved a **23% improvement in bandwidth reservation** during peak times. This new solution was later presented in a segment of Juniper's Technology Conference.
- On-Box Feature Implementation: Merged the predictive model as a customer-facing on-box feature, improving Juniper Networks' product offering and providing a cutting-edge tool for network optimization.
- Advanced Machine Learning Techniques: Gained experience by researching time series forecasting models in machine learning. Utilized ARIMA, SARIMA, and LSTM models to deliver adaptive bandwidth adjustments, aligning with customer needs and outperforming traditional reactive based methods

Advantest Inc., Santa Jose, CA — Software Engineer Intern

May 2019 - August 2019

- Developed an Advanced Test File Detection and Analysis Interface purely in java; which consisted of a sophisticated graphical user interface (GUI) capable of reading program data and conducting scans to identify specific testing file types. The backbone of the features were automation scripts that can detect the file type used by test engineers and produce fast and reliable templates for execution.
 - Successfully incorporated the tool with the company's IDE, enabling widespread adoption by test engineers, resulting in improved testing efficiency across the team. Given the award for Highest Impact Intern.
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Personal Projects

Hack the Box: Security Learning

January 2025 - Current

- Working on the Penetration Tester Path in Hack the Box Academy - Web Security, Network Exploitation, and Privilege Escalation.
- Developing custom enumeration scripts in Python for reconnaissance.

Red-Eye Password Manager

November 2024 - Current

- Developing a secure, local password manager in Rust with features like credential storage, password generation, and encryption using modern cryptographic libraries.
 - Implementing a user-friendly interface using React, enabling seamless password access and management while maintaining strong security principles such as salting and encryption-at-rest.
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Certifications

CompTIA Security +

Current

CSA Certified Cloud Security Knowledge (CCSK)

Current

Offensive Security Certified Professional (OSCP)

In-Progress

EDUCATION

Georgia Institute of Technology - Master of Science in Computer Science

Current

Arizona State University, Tempe - Bachelor of Science in Computer Science